

PRACTICAL QUALITY MANAGEMENT IN THE MEDICAL LABORATORY

A Day-to-Day Guide for Quality Control,
Quality Assurance and
Laboratory Quality Systems



**Practical handbook for
routine laboratory work**

Daily workflow • QC • EQA • CAPA • Risk management
Measurement uncertainty • Inter-instrument variation



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This handbook is intended for education, training and laboratory quality-management support. It does not replace current ISO, CLSI, WHO, national regulatory requirements, manufacturer instructions, laboratory-approved SOPs, accreditation requirements or professional judgment. Numerical examples, QC rules, acceptance limits and measurement-uncertainty examples must be verified against the laboratory's validated method and applicable standards before routine use.

Printed copies should be checked against the current approved edition before use as a laboratory procedure.

About the Author

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The author has prepared this handbook as a practical working guide for medical laboratory professionals who need to connect quality principles with routine bench practice. The emphasis is on daily workflow, internal quality control, troubleshooting, EQA, measurement uncertainty, inter-instrument comparison, CAPA, risk management and usable checklists rather than theory alone.

How to Use This Handbook

- Use the daily workflow and checklists as bench-side reminders.
- Use the IQC and QC-failure chapters when reviewing shifts, trends or rejected control runs.
- Use EQA, inter-instrument comparison and measurement uncertainty chapters for periodic quality review.
- Use CAPA and risk-management sections to document and prevent recurring problems.
- Adapt forms and acceptance criteria to the laboratory-approved QMS before implementation.

Preface

Quality in a medical laboratory is not achieved by running control material once a day. It is created by a connected system: correct patient identification, suitable specimens, functioning equipment, stable reagents, valid calibration, acceptable internal QC, competent staff, reliable result review, timely communication, corrective action and continuous monitoring.

The practical question behind this handbook is simple: What should a laboratory team actually do every day to keep results reliable? The chapters therefore follow the real laboratory workflow—from opening the laboratory in the morning to reviewing quality performance at the end of the month.

Special emphasis is given to internal quality control, quality indicators, QC troubleshooting, inter-instrument variation, measurement uncertainty, risk management and ready-to-use checklists. The goal is to make quality management visible in routine work rather than keeping it only in policies and audit files.

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Quick Reference: Essential QMS Terminology

Use this section as a bench-side vocabulary guide. The definitions are intentionally practical; exact metrological or accreditation wording should follow the laboratory's adopted standards and policies.

Laboratory Quality System at a Glance

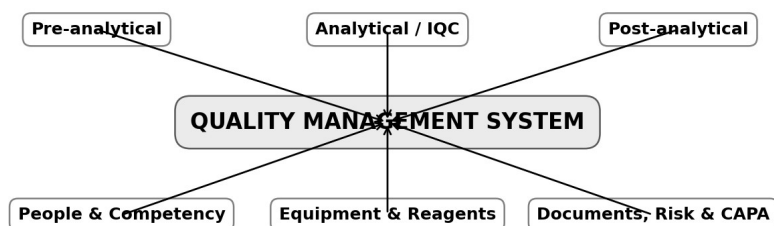


Figure: A quality management system connects the total testing process with people, equipment, documents, risk and improvement.

Quick terminology reminder

- Use laboratory-approved definitions when writing formal policies or accreditation documents.
- For daily teaching, connect every term to an actual laboratory action or example.
- Terms such as TEa, allowable bias, critical values and acceptance limits must come from an approved local or adopted specification - not from an arbitrary universal number.

Term	Simple practical meaning	Laboratory example
Quality	Fitness and reliability of the laboratory service for its intended clinical use.	Correct specimen + valid measurement + timely, safe report.
Quality Control (QC)	Operational techniques used to monitor and control analytical performance.	Run control material and review the LJ chart before patient-result release.

Term	Simple practical meaning	Laboratory example
Quality Assurance (QA)	Planned activities that prevent and detect problems across the total testing process.	Sample-rejection monitoring, competency, report review and audits.
Quality Management System (QMS)	The organized system linking people, processes, equipment, documents, risk and improvement.	Policies, SOPs, QC, EQA, CAPA, training and management review working together.
Accuracy	Closeness of a measurement result to the accepted reference or true value; influenced by both systematic and random effects.	A glucose result is close to the assigned/reference value and consistently reproducible.
Trueness	Closeness of the mean of repeated measurements to a reference value.	Mean QC = 101 mg/dL when reference value = 100 mg/dL.
Precision	Closeness of agreement among repeated measurements.	Repeated QC values cluster closely together.
Repeatability	Precision under the same conditions over a short time.	Same analyzer, operator, reagent lot and run.
Reproducibility / Intermediate precision	Precision when conditions vary within or between laboratories.	Different days/operators/instruments give sufficiently comparable results.
Bias	Systematic difference between the observed mean and a target/reference value.	Target 100, observed mean 103 -> positive bias of 3%.
Mean	Average of a set of results.	Sum of QC results divided by number of results.
Standard Deviation (SD)	Measure of spread around the mean in the same units as the result.	Small SD means QC results are tightly clustered.
Coefficient of Variation (CV%)	SD expressed as a percentage of the mean; a common measure of imprecision.	Mean 100, SD 2 -> CV = 2%.
Total Allowable Error (TEa)	Maximum analytical error allowed for the intended use, based on an adopted performance specification.	Used with bias and CV in a sigma-metric calculation when the laboratory has an approved TEa source.

Term	Simple practical meaning	Laboratory example
Allowable Bias	Maximum systematic difference accepted by the laboratory for a method or comparison.	New reagent lot is accepted only if observed bias is within the predefined limit.
Measurement Uncertainty (MU)	Quantified doubt around a measurement result under stated conditions.	Creatinine 1.00 ± 0.06 mg/dL under the laboratory's MU model.
Traceability	Property linking a result to a reference through a documented calibration chain.	Calibrator value is traceable to an accepted reference system where available.
Calibration	Establishing the relationship between instrument response and analyte concentration.	Analyzer response is aligned using a manufacturer calibrator.
Calibration Verification	Confirming that the calibration remains valid across the relevant measuring range.	Multiple levels recover within predefined limits.
Method Verification	Confirming that an established method performs acceptably in the local laboratory.	Verify precision, bias/range and other required characteristics before routine use.
Validation	Providing objective evidence that a process or method is fit for its intended use, especially when developed or modified locally.	Validate an in-house control preparation process before routine use.
Reference Interval	Range expected for a defined reference population, commonly the central 95% for many analytes.	Use age/sex/method-appropriate intervals in the report.
Analytical Measurement Range (AMR)	Range over which a method can directly measure without dilution or special treatment, as verified.	Patient result above AMR requires validated dilution/reanalysis.
Internal Quality Control (IQC)	Routine control material used to monitor day-to-day analytical stability.	Two QC levels are run and reviewed before release of patient testing.
EQA / Proficiency Testing	External assessment comparing laboratory performance with assigned values or peer groups.	Unacceptable EQA result triggers investigation and CAPA.

Term	Simple practical meaning	Laboratory example
Lot-to-Lot Verification	Comparison performed before routine use of a new reagent/control lot.	Patient samples are tested on old and new lots and differences are assessed.
Inter-Instrument Comparison	Assessment of comparability between analyzers measuring the same analyte.	Monthly paired-patient comparison between analyzer A and B.
Nonconformity (NC)	Failure to meet a requirement.	Critical value was not communicated within the approved time.
Correction	Immediate action to fix a detected problem.	Recollect a mislabeled specimen.
Corrective Action	Action to eliminate the cause of a detected nonconformity and reduce recurrence.	Revise identification workflow and retrain staff after recurrent labeling errors.
CAPA	Structured corrective and preventive/risk-reduction actions with follow-up for effectiveness.	Root cause -> action -> owner -> due date -> effectiveness check.
Root Cause Analysis (RCA)	Systematic investigation to identify why a problem occurred.	Use 5 Why or fishbone analysis for repeated hemolysis.
Quality Indicator	Measurable variable used to monitor process performance over time.	Sample rejection rate, TAT compliance, QC failure rate.
Risk	Combination of the likelihood of an event and the severity of its consequence.	Wrong-patient labeling is low frequency but high severity.
Critical Value	Laboratory result indicating potentially life-threatening risk that requires urgent communication under policy.	Critical potassium is communicated and documented promptly.
Delta Check	Comparison of a current result with a previous result to identify an unexpected change.	Hb falls from 13.8 to 8.2 g/dL -> investigate true change or error.
Turnaround Time (TAT)	Time interval defined by the laboratory for completion of a testing process.	Monitor percentage of emergency results released within target.

Term	Simple practical meaning	Laboratory example
SOP	Controlled written instructions describing how a procedure is performed.	Current approved version must be available at the bench.
Document Control	System for approval, revision, distribution and withdrawal of documents.	Obsolete SOPs are removed from use and archived.
Competency	Demonstrated ability of staff to perform assigned work correctly.	Direct observation, QC review, sample testing and problem-solving assessment.

Chapter 1: What is Quality in the Laboratory?

Laboratory quality means producing a result that is sufficiently accurate, reliable, timely, traceable and clinically useful for the intended patient-care decision. A high-quality laboratory does not only produce a number; it controls the complete path from test request to clinical use of the report.

Quality, QC, QA and QMS: the practical difference

Term	Practical meaning	Example in the laboratory
Quality	The overall fitness and reliability of the laboratory service.	A potassium result is correct, released on time and communicated urgently when critical.
QC (Quality Control)	Operational techniques used to detect analytical problems.	Running glucose control and reviewing the LJ chart before releasing patient glucose results.
QA (Quality Assurance)	Planned activities that prevent errors across the testing process.	Training phlebotomists, monitoring sample rejection and validating reports.
QMS (Quality Management System)	The organized system that connects people, processes, documents, equipment, risk and improvement.	Policies, SOPs, audits, QC, EQA, CAPA, competency, document control and management review working together.

A simple practical example

Suppose a patient's serum potassium is reported as 7.2 mmol/L. Quality requires more than seeing that the number is high.

Confirm the specimen belongs to the correct patient.

Check whether the sample is hemolyzed or contaminated.

Confirm analyzer and IQC status.

Repeat/verify according to policy when appropriate.

Assess whether the value meets the laboratory critical-value threshold.

Communicate the critical result to an authorized clinician and document read-back.

Record any nonconformity if the result was delayed or the specimen was unsuitable.

Core message

- **QC is part of quality, but QC alone is not the complete quality system.**
- **A good QMS connects pre-analytical, analytical and post-analytical control with people, equipment and documentation.**

Chapter 2: Types of Quality Control

Different quality-control activities answer different questions. Internal QC mainly asks, “Is the analytical system stable today?” EQA asks, “How does our result compare with an external target or peer group?” Inter-laboratory and inter-instrument comparisons ask, “Are different testing locations or analyzers producing sufficiently comparable results?”

QC activity	Main purpose	Typical frequency	Practical evidence
Internal QC (IQC)	Detect day-to-day analytical instability, shift, trend or imprecision.	Daily / per run / as risk requires	Control results, LJ chart, rule evaluation, corrective action
External Quality Assessment (EQA/PT)	Assess performance against external assigned values or peers.	Program schedule	EQA report, score, root-cause analysis
Inter-Laboratory Comparison (ILC)	Compare results between laboratories when EQA is unavailable or as an additional check.	Planned periodic	Shared-sample comparison, bias summary
In-house control	Provide locally prepared/stabilized QC material when commercial controls are unavailable or impractical.	As validated	Preparation record, homogeneity/stability data, assigned mean/SD
Lot-to-lot verification	Ensure new reagent/control lot performs comparably to current lot.	At lot change	Parallel comparison and acceptance decision
Inter-instrument comparison	Confirm multiple instruments measuring the same analyte are comparable.	Monthly/quarterly or risk-based	Patient-sample difference/bias report

When each QC type is most useful

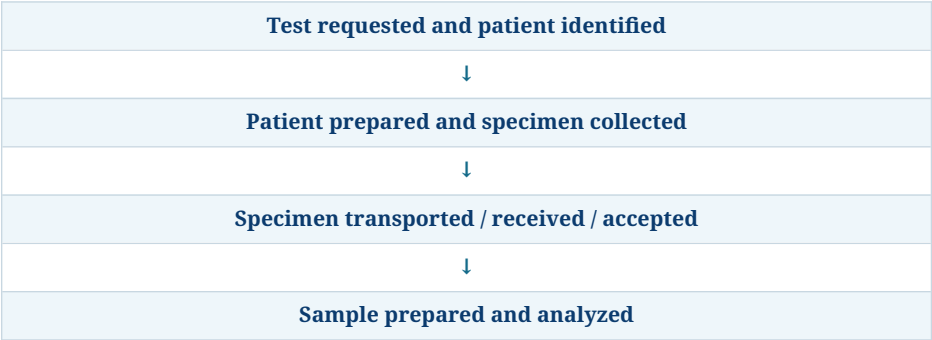
- IQC is the first-line daily safety check before or during patient testing.
- EQA is not a replacement for IQC; it provides an independent external performance check.
- ILC is useful for methods/analytes not covered by EQA, but the comparison design must be documented.
- In-house controls require stability, homogeneity and target assignment before routine use.
- Lot comparison is preventive control: detect a problem before the new lot affects patient results.
- Inter-instrument comparison is essential where the same patient may be tested on different analyzers.

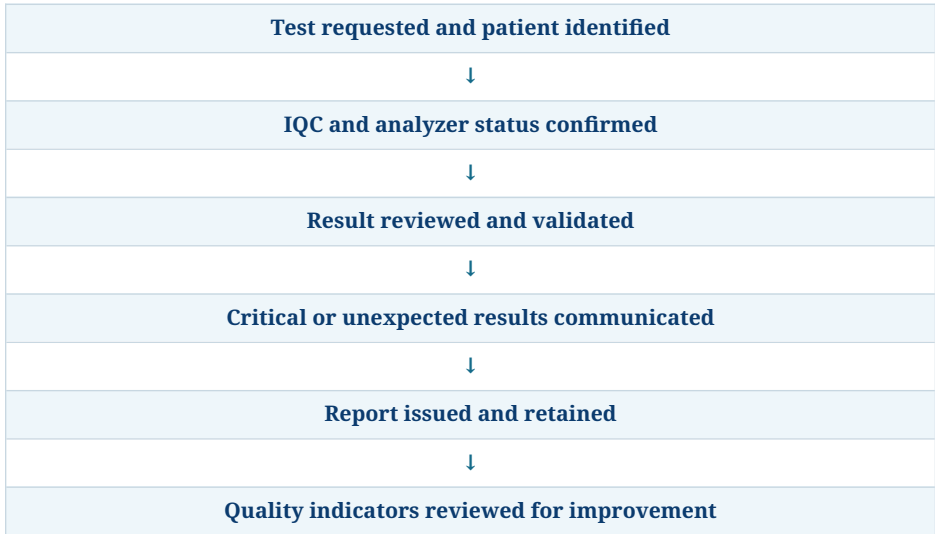
Chapter 3: Different Phases of Laboratory Quality

Errors can occur before, during and after measurement. Good laboratories build controls in all three phases.

Phase	Common errors	Daily action	Preventive control
Pre-analytical	Wrong patient; wrong tube; hemolysis; clot; insufficient volume; delayed transport; wrong storage	Check identifiers, specimen condition, collection time, transport and rejection criteria	Collection SOP, staff training, rejection monitoring, transport limits
Analytical	Expired reagent; calibration drift; pipetting fault; control failure; instrument error; interference	Check maintenance, calibration status, reagent/control, IQC and analyzer flags	Preventive maintenance, IQC, calibration verification, lot verification
Post-analytical	Transcription error; delayed report; missed critical value; incorrect reference interval; wrong amended report	Verify results, flags, delta checks, critical values, report authorization	LIS rules, review SOP, critical-value policy, TAT indicators, amended-report control

Total testing process flow





Daily question for every phase

- Pre-analytical: Is this the right, acceptable specimen?
- Analytical: Is the measurement system under control?
- Post-analytical: Is this result valid, timely and safely communicated?

Chapter 4: Types of Laboratory Errors

Understanding error type helps choose the correct investigation. Repeating QC without understanding whether the problem is random, systematic or gross often wastes time.

Error type	Meaning	Typical sign	Practical example	Likely action
Random error	Unpredictable variation affecting precision	Scattered QC results; isolated rule violation	Air bubble, intermittent pipetting issue	Repeat, inspect mechanics, improve precision
Systematic error	Consistent bias affecting accuracy	Shift or trend in QC; consistent patient bias	Reagent deterioration, calibration drift	Check calibration/reagent/lot , compare method
Gross error	Major avoidable mistake	Result incompatible with process or patient	Wrong sample, decimal error, wrong reagent	Stop reporting, correct immediate cause, document NC/CAPA

Phase-based error examples

Phase	Example	How it may appear	Control
Pre-analytical	EDTA contamination of chemistry sample	High K + low Ca with unexpected pattern	Collection sequence/training; sample review
Pre-analytical	Hemolysis	Falsely high K, LDH, AST depending on method	Hemolysis index; recollection criteria
Analytical	Calibration drift	Multiple QC points shift to one side of mean	Calibration review and corrective action
Analytical	Probe partially blocked	Imprecision across several tests	Maintenance, precision check

Phase	Example	How it may appear	Control
Post-analytical	Critical result not communicated	Correct number but unsafe service	Critical-call log and indicator
Post-analytical	Wrong unit/reference interval	Clinical misinterpretation	LIS master verification

Do not confuse detection with cause

- A Westgard-rule alert is a signal, not a root cause.
- The root cause may be reagent, calibration, instrument, environment, control material, operator or sample handling.

Chapter 5: Quality Indicators

Quality indicators convert routine events into measurable evidence of performance. A useful indicator has a clear numerator, denominator, target, frequency and action threshold.

General percentage indicator: Indicator (%) = (Number of events / Total opportunities) × 100

Example: If 18 of 1,200 samples are rejected: rejection rate = $18/1200 \times 100 = 1.5\%$.

Indicator	Suggested formula	Why it matters
Sample rejection rate	Rejected specimens / total received × 100	Shows pre-analytical quality
Hemolysis rate	Hemolyzed specimens / applicable specimens × 100	Tracks collection/transport quality
TAT compliance	Results within target / total results × 100	Measures timeliness
Critical value communication	Critical results communicated within target / total critical results × 100	Patient-safety indicator
Corrected report rate	Corrected reports / total reports × 100	Shows post-analytical reliability
QC failure rate	Rejected QC runs / total QC runs × 100	Shows analytical stability
EQA satisfactory performance	Acceptable EQA results / total EQA results × 100	External performance
Equipment downtime	Downtime hours / operating hours × 100	Service continuity

What makes an indicator useful?

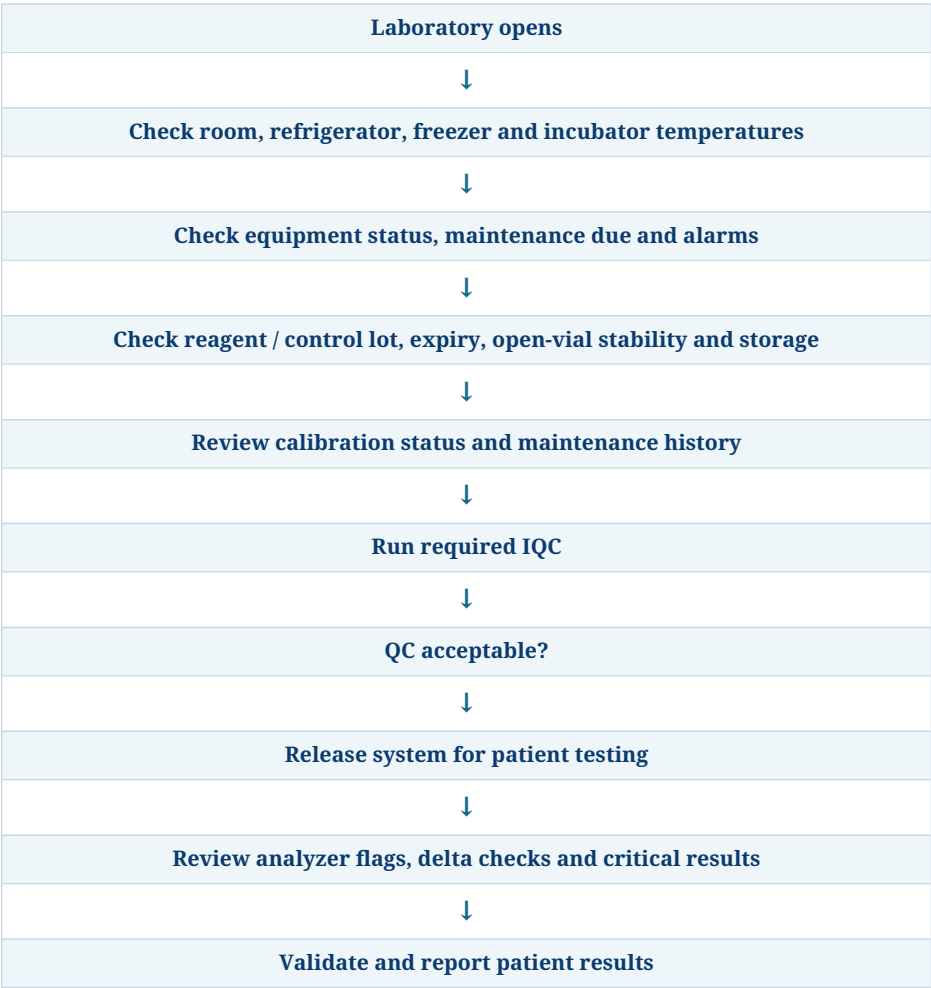
- Definition is unambiguous.
- Data source is available and reliable.
- Target is realistic and approved.

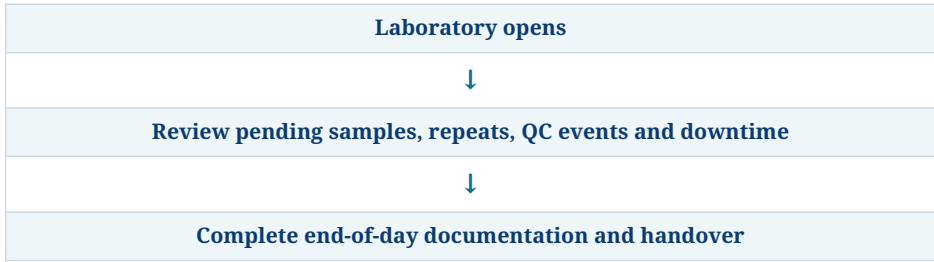
- Trend is reviewed, not just a single month.
- Poor performance triggers documented action.
- Indicator is retired or revised if it no longer drives improvement.

Chapter 6: Daily Laboratory Quality Workflow

This chapter is the operational heart of the handbook. The safest laboratory day begins with readiness checks, not with the first patient sample.

Start-to-end daily quality flow





Start-of-day readiness checklist

Check	Acceptable?	Action if not acceptable
Room and storage temperatures	Within approved limits	Assess exposure duration; quarantine affected materials if needed
Analyzer startup / self-check	No unresolved alarm	Troubleshoot before patient testing
Water / wash solution / waste	Adequate and within specification	Refill/replace; inspect leaks
Reagents and calibrators	Correct lot, expiry, storage	Replace or document validated exception
Controls	Correct level, lot, expiry, preparation	Prepare correctly; do not use compromised control
Calibration	Current and valid	Calibrate only when indicated by method/procedure
IQC	Acceptable by laboratory rule	Do not release affected patient results until resolved

During patient testing

- Watch analyzer alarms, flags, blank/reaction errors and carryover warnings.
- Do not ignore a result only because it is technically within the analytical measuring range.
- Use delta checks or clinical plausibility review for unexpected changes.
- Repeat testing should follow a defined policy; avoid unnecessary repeats that delay care.
- Critical results must be communicated and documented according to the approved policy.

End-of-day review

- Confirm all pending specimens are resolved or handed over.
- Review failed controls, recalibrations, repeated patient results and amended reports.
- Check reagent remaining volume and next-day requirements.
- Complete maintenance, temperature and cleaning records.
- Document unresolved issues for the next shift.

Chapter 7: How to Run Daily IQC on the Instrument

Daily IQC should be handled as a controlled analytical procedure, not as a quick ritual before patient testing. Correct storage, thawing or equilibration, mixing, lot selection, analyzer setup, result review and documentation all influence the reliability of the QC result.

The exact preparation, handling and stability instructions must always follow the control manufacturer's instructions and the laboratory-approved SOP. The practical sequence below can be adapted to chemistry, immunoassay, hematology and other quantitative systems.

Before removing the control from storage

- Confirm which tests and QC levels are due for the current run or shift.
- Check the control material name, level, lot number and expiry date.
- Check the manufacturer-defined storage conditions and the open-vial or reconstituted stability.
- Confirm that the analyzer has the correct active QC lot and the assigned target mean, SD or acceptable range.
- Review whether the reagent lot, calibration status or major maintenance has changed since the previous acceptable QC.

Storage, thawing and equilibration

Control condition	Practical handling	Common mistake
Refrigerated liquid control	Equilibrate to the manufacturer-specified working temperature before use. Mix gently and thoroughly as instructed. Do not warm to 37 °C unless the product insert or a validated laboratory procedure specifically requires it.	Running the control too cold, warming it unnecessarily, or mixing it incorrectly.
Frozen control	Thaw completely using the manufacturer-approved procedure. Mix gently after complete thawing and allow the material to reach the specified working temperature before analysis. Avoid repeated freeze-thaw cycles unless validated.	Partial thawing, repeated freeze-thaw cycles, or using the material before it is homogeneous.

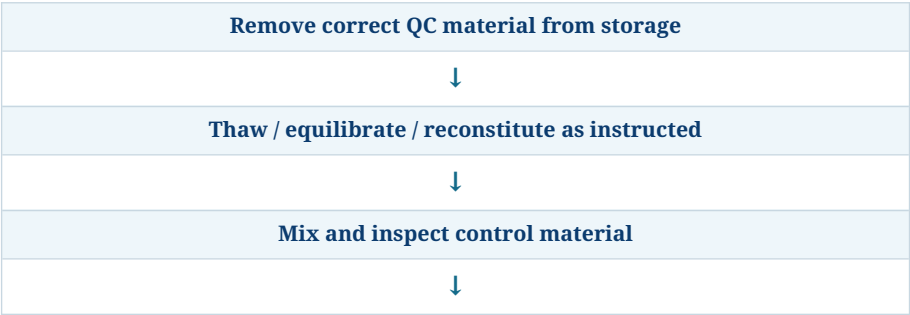
Control condition	Practical handling	Common mistake
Lyophilized control	Reconstitute with the exact diluent type and volume. Observe the specified standing time, then mix gently until completely dissolved.	Wrong diluent or volume, vigorous shaking, incomplete dissolution, or incorrect standing time.
Whole-blood hematology control (commonly stored at 2-8 °C; follow IFU)	Store at 2-8 °C unless otherwise specified. Bring to the manufacturer-recommended working temperature and mix exactly as instructed, commonly by gentle inversion or roller mixing. Do not incubate at 37 °C unless specifically instructed.	Insufficient mixing, cell settling, excessive agitation, or warming outside the validated procedure.

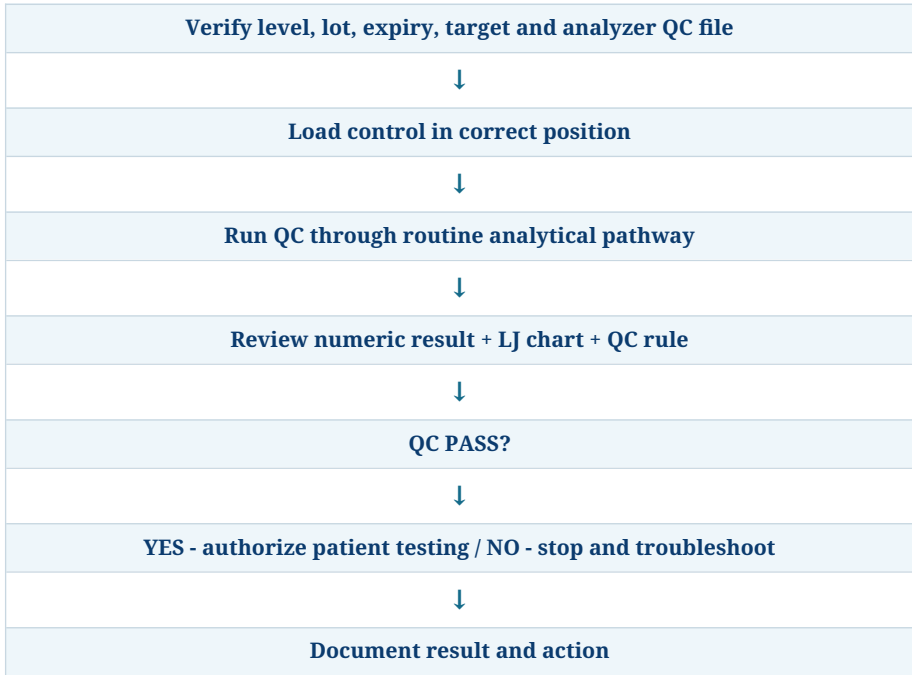
Do not assume that every QC material requires the same thawing, equilibration or mixing procedure. Follow the product insert, manufacturer instructions and the laboratory-validated procedure.

Analyzer setup before running QC

- Open the QC menu or control file on the instrument.
- Verify control material, level and lot number.
- Verify expiry date and active status.
- Verify the target mean, SD or manufacturer/laboratory-defined acceptable range.
- Confirm reagent lot and calibration status.
- Check analyzer readiness: there should be no unresolved alarms, overdue maintenance or probe/wash errors.
- Assign or load the control in the correct rack, cup, position or barcode location.

Daily IQC operational flow





After the control has run

- Review the actual QC value, not only the analyzer PASS/FAIL symbol.
- Check the Levey-Jennings chart for shift, trend or unusual movement even when the current point is within range.
- Review all QC levels together; one level may reveal concentration-dependent problems.
- Confirm the control result belongs to the correct lot and level.
- If QC passes, document or electronically authorize the QC run according to the SOP.
- If QC fails, do not release affected patient results until the failure is investigated and acceptable QC is restored.

What to do with the control after use

- Return the control to storage only if the manufacturer permits reuse and the validated stability period remains valid.
- Record the opened or reconstituted date where required.
- Do not refreeze unless the manufacturer or validated in-house procedure allows it.

Discard control if contaminated, expired, unstable or otherwise compromised.

High-yield daily IQC points

- Correct storage and preparation are part of IQC quality.
- Verify the active QC lot and assigned target at every lot change.
- A QC value inside range can still show a meaningful trend or shift.
- Never repeat failed QC again and again simply to obtain one passing result.
- Release patient testing only after the analytical system is demonstrated to be under control.

Daily IQC bench checklist

Check	Done
Correct QC material / level / lot	☐
Expiry and stability valid	☐
Thawed/equilibrated/ reconstituted correctly	☐
Mixed and visually acceptable	☐
Analyzer QC lot/target verified	☐
Reagent/calibration status acceptable	☐
QC run completed	☐
LJ chart and rules reviewed	☐
Pass/fail decision documented	☐

Worked Example: Daily Chemistry IQC Run

Illustrative example only. The actual control preparation, QC frequency, target values, acceptance rules and release criteria must follow the manufacturer instructions and the laboratory-approved SOP.

Scenario

A chemistry analyzer is ready for the morning run. Two levels of glucose control are scheduled before routine patient testing. The reagent lot is unchanged, calibration is valid, and no unresolved analyzer alarms are present.

Step-by-step bench sequence

- Remove the correct Level 1 and Level 2 control material from validated storage and verify the material name, lot number, expiry date and stability.
- Equilibrate, thaw or reconstitute each control exactly as instructed. Mix gently until homogeneous and inspect for precipitate, clot, contamination or other abnormal appearance.
- Open the analyzer QC menu and confirm that the active QC lot, assigned mean, SD or acceptable range correspond to the control in hand.
- Confirm reagent lot, calibration status and analyzer readiness, then load each QC level in the correct rack/cup/barcode position.
- Run both QC levels through the routine analytical pathway.
- Review the numeric result, SD position or laboratory-approved rule, and the Levey-Jennings trend. Do not rely only on a green PASS symbol.
- If both levels are acceptable and no trend/shift warning requires action, authorize patient testing according to the SOP.
- If either level is unacceptable, stop release of affected patient results and follow the QC-failure troubleshooting procedure.

Illustrative QC review

QC level	Assigned mean	SD	Observed result	SD position	Decision
Level 1	100 mg/dL	2 mg/dL	101.5 mg/dL	+0.75 SD	Acceptable in this illustrative example
Level 2	250 mg/dL	5 mg/dL	254 mg/dL	+0.80 SD	Acceptable in this illustrative example

Interpretation: Both results are close to their assigned means and do not trigger a rejection condition in this illustrative scenario. The laboratory should still review the LJ chart for shift or trend before releasing the analytical run.

Practical reminder

- A single passing result does not justify ignoring a persistent trend.
- Do not change the target mean/SD or repeatedly rerun failed QC simply to obtain a pass.
- Document the QC result, review and any corrective action before patient-result release.

Chapter 8: Internal Quality Control in Daily Practice

IQC monitors analytical stability using control material with known or assigned expected performance. The goal is not to make the control value equal the mean every day; the goal is to detect statistically or clinically important changes before patient results are affected.

Core statistics

Mean: $\text{Mean} = \Sigma x / n$

Example: Control results: 100, 101, 99, 100, 102 → mean = 100.4

Standard deviation (SD): $\text{SD} = \sqrt{[\Sigma(x - \text{mean})^2 / (n - 1)]}$

Example: SD describes dispersion around the mean.

Coefficient of variation: $\text{CV}\% = (\text{SD} / \text{Mean}) \times 100$

Example: Mean 100, SD 2 → CV = 2.0%.

Bias: $\text{Bias}\% = [(\text{Observed mean} - \text{Target value}) / \text{Target value}] \times 100$

Example: Observed mean 102, target 100 → bias = +2%.

Levey-Jennings chart interpretation

- Plot control values against time with mean and ± 1 , ± 2 and ± 3 SD lines.
- Look for isolated excursions, shifts and gradual trends.
- Assess each control level independently and together.
- Use laboratory-approved QC rules and clinical risk; do not apply rules mechanically without understanding method behavior.

Signal	Possible meaning	Immediate thought
Single point beyond 3 SD	Strong outlier / random or major event	Stop and investigate
Two consecutive results beyond same 2 SD	Possible systematic shift	Reagent/calibration/lot

Signal	Possible meaning	Immediate thought
Large difference between two levels	Possible random error	Pipetting/mechanical issue
Several consecutive results on one side of mean	Shift	Calibration or reagent change
Gradual movement in one direction	Trend	Reagent aging, temperature, lamp/probe deterioration

From Raw QC Data to a Levey-Jennings Chart - Step by Step

A Levey-Jennings (LJ) chart becomes much easier to interpret when the reader first understands how the mean and SD lines are created from stable QC data. The following worked example uses illustrative glucose-control results.

Run	Result mg/dL	Run	Result mg/dL
1	99	11	102
2	101	12	101
3	100	13	99
4	102	14	100
5	98	15	104
6	100	16	103
7	101	17	102
8	99	18	101
9	103	19	100
10	100	20	99

Worked calculation

- Mean = sum of 20 results / 20 = 100.7 mg/dL.
- SD (sample SD) = approximately 1.59 mg/dL.
- CV% = $(1.59 / 100.7) \times 100$ = approximately 1.58%.
- +1 SD = 102.3; +2 SD = 103.9; +3 SD = 105.5 mg/dL.
- -1 SD = 99.1; -2 SD = 97.5; -3 SD = 95.9 mg/dL.

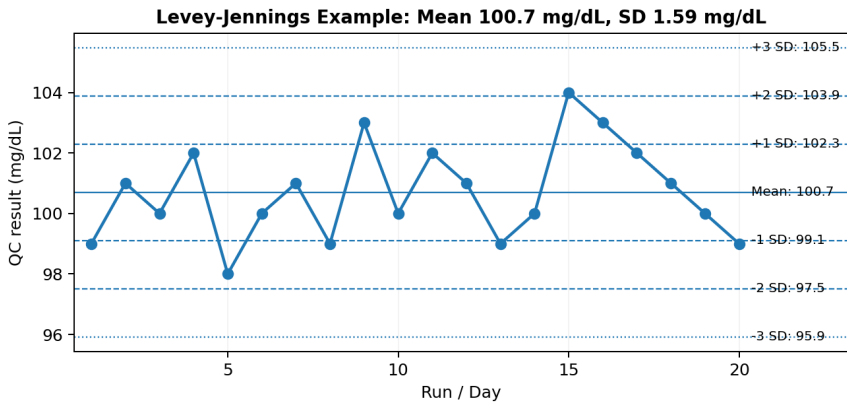


Figure: Worked Levey-Jennings chart created from 20 illustrative QC results.

How to Read an LJ Chart

- Review the current point in relation to the mean and SD limits.
- Look backward: a current point can be inside the limits while a shift or trend is developing.
- Review all QC levels together because concentration-dependent problems may affect one level before another.
- Distinguish an isolated random excursion from a persistent systematic pattern.
- Apply the laboratory-approved QC rule set; do not reject or accept a run solely from the chart appearance without the defined rule and clinical risk context.

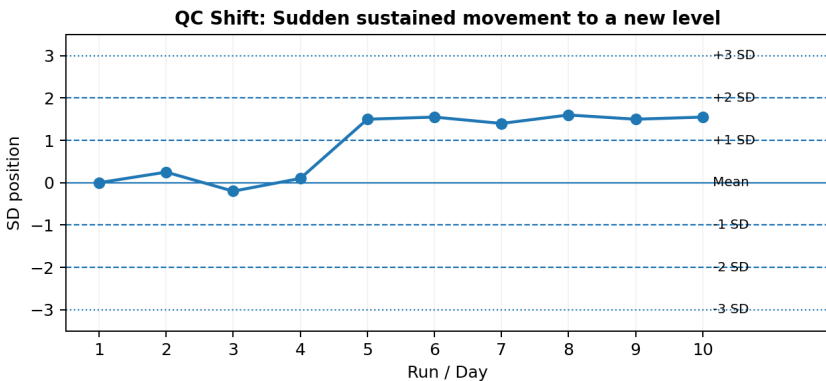


Figure: A QC shift is a sudden sustained movement to a new level.

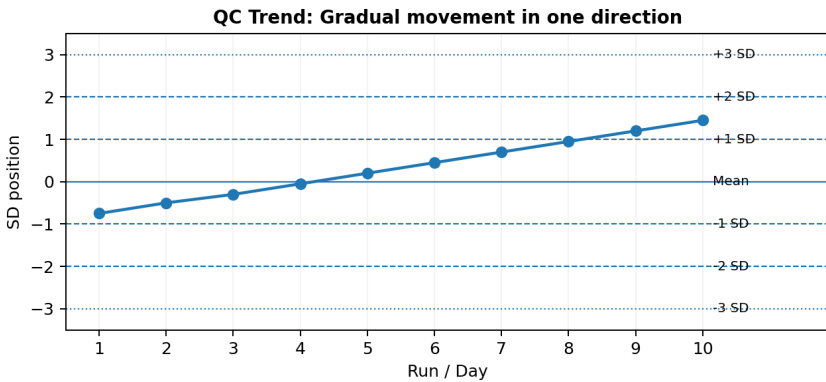


Figure: A QC trend is gradual movement in one direction over time.

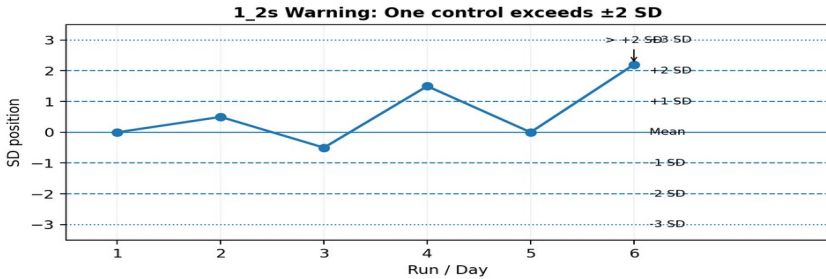
Westgard Multirule QC - Practical Interpretation

Westgard multirules combine statistical signals to improve error detection while limiting unnecessary rejection. The classic rules below are teaching examples. The actual rule set, number of control levels and frequency must be selected for the method's performance and clinical risk.

Rule	Signal	What it suggests	Practical response
1_2s	One control exceeds ± 2 SD	Warning in classic multirule use	Review other rules and context; not an automatic rejection by itself.
1_3s	One control exceeds ± 3 SD	Strong random/major error signal	Reject affected run and investigate.
2_2s	Two consecutive results exceed same ± 2 SD	Systematic error signal	Reject; review calibration, reagent, lot and systematic causes.
R_4s	Difference between two controls is ≥ 4 SD within a run	Random error signal	Reject; check pipetting, mixing, bubbles, probe/mechanics.
4_1s	Four consecutive results exceed same ± 1 SD	Systematic error signal	Reject/investigate according to approved multirule design.
10x	Ten consecutive results on same side of mean	Shift / systematic bias signal	Investigate persistent bias, calibration or reagent-related change.

1_2s - Warning Rule

A single control result crosses $+2$ SD or -2 SD. In the classic multirule approach it is a warning signal, not an automatic reject rule. Check the other control level, recent trend and whether another reject rule is also triggered.



Illustrative 1_2s pattern.

1_3s - Reject Rule

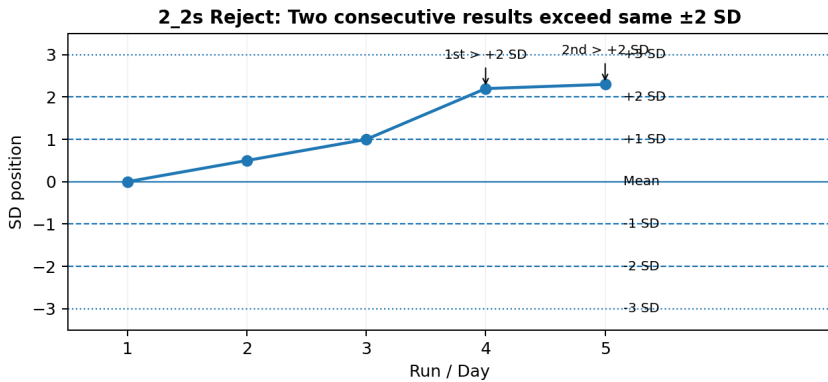
One control result exceeds $+3$ SD or -3 SD. This is unlikely from ordinary stable variation and should trigger rejection and investigation of a major/random event or other analytical problem.



Illustrative 1_3s pattern.

2_2s - Systematic Error Rule

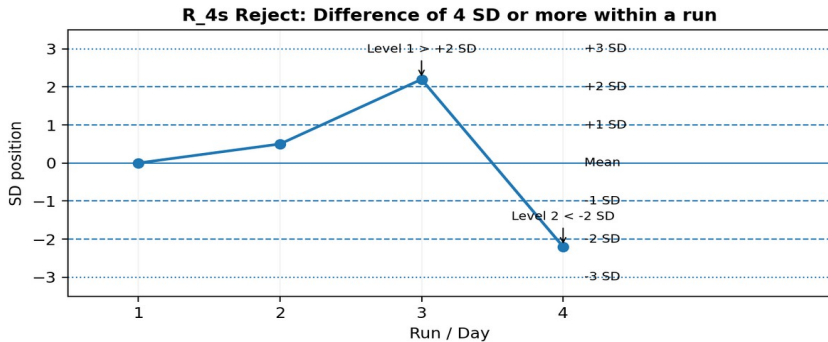
Two consecutive control results exceed the same $+2$ SD or -2 SD limit. The repeated movement in one direction suggests a systematic change such as calibration shift, reagent/lot bias or deterioration.



Illustrative 2₂s pattern.

R₄s - Random Error Rule

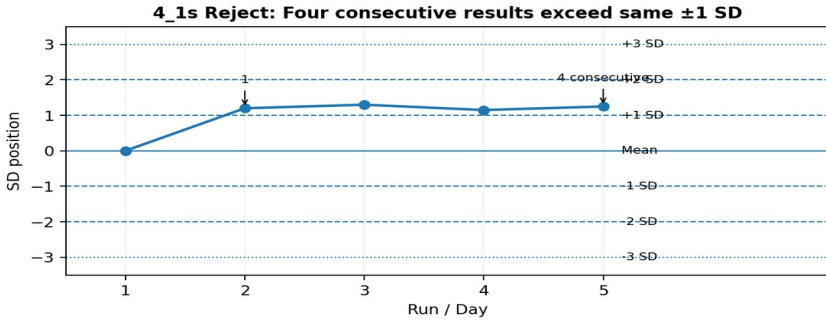
Within the same analytical run, one control is above $+2$ SD and another is below -2 SD, creating a separation of at least 4 SD. This pattern suggests excessive random variation such as pipetting, bubble, mixing or intermittent mechanical error.



Illustrative R₄s pattern.

4_1s - Systematic Error Rule

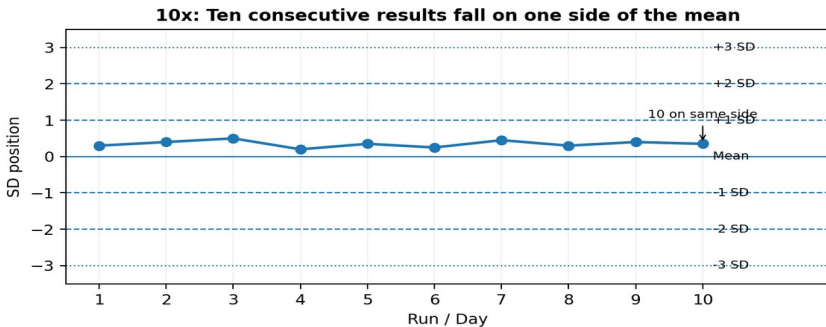
Four consecutive control observations exceed the same +1 SD or -1 SD limit. This repeated one-sided behavior suggests a small but persistent systematic shift.



Illustrative 4_1s pattern.

10x - Shift Rule

Ten consecutive control results fall on the same side of the mean. The individual points may still be within ± 2 SD, but the persistent one-sided pattern suggests systematic bias or a new analytical level.



Illustrative 10x pattern.

Westgard interpretation discipline

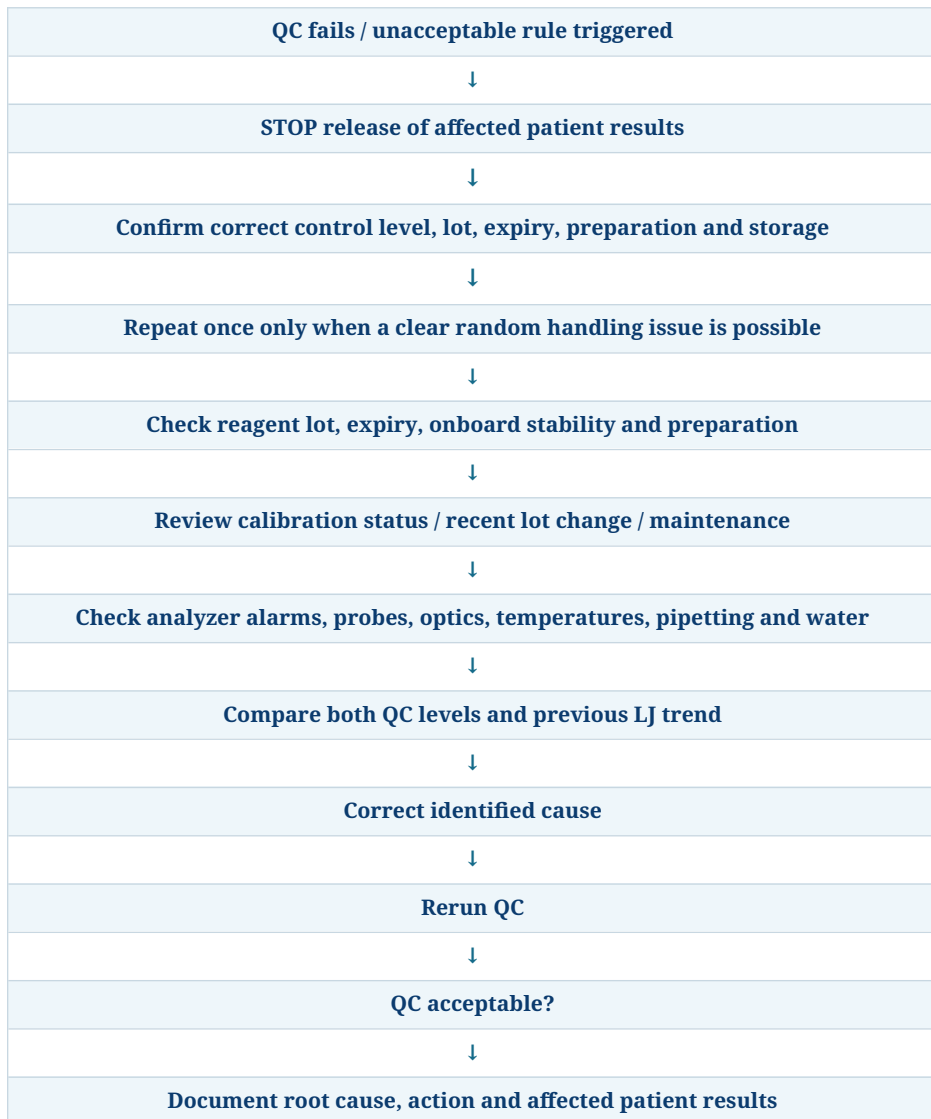
- A rule violation is a signal that the analytical process may be unstable; it is not itself the root cause.
- Do not repeatedly rerun failed controls until one point happens to pass.
- Link random-error rules to imprecision/mechanical checks and systematic-error rules to calibration/reagent/lot/trend checks.
- Use rule sets appropriate to the assay sigma/risk profile rather than applying every rule identically to every test.

Accept / warn / reject decision

The laboratory should define which rules are warning rules and which are rejection rules, based on assay performance, QC frequency, clinical risk and manufacturer/validation evidence.

Chapter 9: What to Do When QC Fails?

QC failure troubleshooting flow



QC fails / unacceptable rule triggered
↓
Release testing only after acceptance

Common QC failure patterns and targeted checks

Pattern	First checks	Do not do
One level fails, other level stable	Control vial/preparation, level-specific issue, range assignment	Immediately recalibrate every assay
Both levels shift same direction	Reagent/calibration/lot/instrument condition	Keep repeating controls indefinitely
Erratic alternating results	Pipetting, bubbles, probe, mixing, intermittent mechanics	Assume target mean is wrong
Gradual trend	Reagent aging, calibration drift, temperature, lamp/probe	Wait until 3 SD failure if trend is clear
After reagent lot change	Lot verification, calibration requirement, new lot bias	Use old target without verification

Patient-result impact assessment

- Identify the time interval since the last acceptable QC.
- Identify affected analyses and patient results.
- Review whether results need repeat testing, correction or clinician notification.
- Document the decision even when no patient result was affected.

Westgard Signal to Troubleshooting - Quick Link

QC signal	Targeted checks	Practical caution
1_{2s} warning	First review other QC levels, previous points and other rules.	Avoid automatic rejection based only on 1 _{2s} in a classic multirule design.
1_{3s}	Control preparation, pipetting, bubbles, probe, intermittent analyzer event; also review gross systematic problems.	Stop affected release and investigate before accepting the run.

QC signal	Targeted checks	Practical caution
R_4s	Pipetting imprecision, poor mixing, air bubble, probe/dispense issue, unstable reaction.	Focus first on random/imprecision sources.
2_2s	Calibration shift, reagent/lot bias, target assignment, environmental or instrument systematic change.	Review systematic causes and recent lot/calibration changes.
4_1s	Small persistent calibration/reagent shift, temperature or gradual analytical change.	Investigate before the shift becomes larger.
10x / sustained one-side shift	New calibration level, lot change, reagent aging, target/mean mismatch, systematic bias.	Compare with event history and patient comparison before redefining target statistics.
Trend	Reagent deterioration, calibration drift, lamp/probe deterioration, temperature or gradual maintenance-related change.	Correct the cause; do not wait for an extreme 3 SD failure if the trend is convincing.